

Static Spherically Symmetric Spacetimes

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Summary

This note presents a self-contained exposition of stationary and static spacetimes in general relativity, with detailed derivations supplementing the standard treatment in Wald (1984). We clarify the equivalence between the definition of stationarity via isometries and the existence of a timelike Killing vector field, and provide a step-by-step proof that staticity is equivalent to the Frobenius condition $\xi_{[a}\nabla_b\xi_{c]} = 0$. The construction of adapted coordinates for static spacetimes is elaborated, including a proof of the orthogonality of ξ^a to each hypersurface Σ_t . The note concludes with the standard form of the metric for static, spherically symmetric spacetimes.

Keywords: Stationary spacetime, Static spacetime, Killing vector field, Frobenius theorem, Spherical symmetry, General relativity

1 Stationary Spacetimes

Definition 1.1 (Stationary Spacetimes). *A spacetime is said to be stationary if there exists a one-parameter group of isometries, ϕ_t , whose orbits are timelike curves.*

The isometry satisfies $\phi_t^* g_{ab} = g_{ab}$. Note that this is an equality of tensor fields, meaning that at each point p , the two tensors $g_{ab}|_p$ and $\phi_t^* g_{ab}|_p$ are equal, i.e.,

$$g_{ab}|_p(v_1)^a(v_2)^b = \phi_t^* g_{ab}|_p(v_1)^a(v_2)^b = g_{ab}|_{\phi_t(p)}(\phi_{t*}v_1)^a(\phi_{t*}v_2)^b, \quad \forall p \in M, v_1, v_2 \in V_p. \quad (1)$$

If we interpret the parameter t as a time translation, then $g_{ab}|_p$ can be understood as the metric at a given spatial point at time t_0 , while $g_{ab}|_{\phi_t(p)}$ is the metric at the same spatial point but at time $t_0 + t$. The vector $(\phi_{t*}v_1)^a$ at time $t + t_0$ is regarded as the vector corresponding to v_1 at time t_0 . Thus, equation (1) implies that for any t , the contraction of the metric with the corresponding vectors at time t_0 is the same as that at time $t_0 + t$. This groups of isometries expresses the “time translation symmetry” of the spacetime.

The diffeomorphism ϕ_t gives a one-parameter group of diffeomorphisms ϕ : given a parameter $t \in \mathbb{R}$, ϕ is determined as ϕ_t . Given $p \in M$, we obtain $\phi_t(p) \in M$, so $\phi : \mathbb{R} \times M \rightarrow M$. For each $p \in M$, the map $\phi(p) : \mathbb{R} \rightarrow M$ is a smooth curve through p , and its tangent vector $v|_p$ at $\phi_p(0)$ defines a smooth vector field v on M . Conversely, let v be a smooth vector field on M . For each $p \in M$, define $\phi_t(p)$ to be the point on the integral curve of v through p whose parameter differs from that of p by t . Then for each $t \in \mathbb{R}$, the integral curves define a diffeomorphism $\phi_t : M \rightarrow M$. From the above discussion, an isometry ϕ_t corresponds to a smooth timelike vector field ξ^a (defined as a Killing vector field). Hence we say that a spacetime is stationary if and only if it admits a timelike Killing vector field ξ^a .

2 Static Spacetimes

Definition 2.1 (Static Spacetimes). *A spacetime is said to be static if it is stationary and if, in addition, there exists a (spacelike) hypersurface Σ which is orthogonal to the orbits of the isometry.*

As a rough example, if matter rotates uniformly so that the overall spacetime structure does not change with time, then the spacetime is stationary; staticity imposes a stronger requirement: not only must the spacetime be time-independent, but the matter itself must be static. However, I have not yet found a way to connect this intuition with the rigorous definitions.

Theorem 2.2. *Definition 2.1 is equivalent to the requirement that the timelike Killing vector field ξ^a satisfy*

$$\xi_{[a} \nabla_b \xi_{c]} = 0. \quad (2)$$

Proof. By Frobenius's theorem, the distribution W orthogonal to ξ^a is integrable if and only if its annihilator T^* , spanned by ξ_a , satisfies

$$d\xi = \mu \wedge \nu, \quad (3)$$

for some $\mu \in T^*$ and some one-form ν . Since T^* is one-dimensional, we may write $\mu = f\xi$, so

$$d\xi = f \xi \wedge \nu. \quad (4)$$

In component form, this is

$$\nabla_{[a} \xi_{b]} = \xi_{[a} \tilde{\nu}_{b]}, \quad (5)$$

where $\tilde{\nu} = f \nu$. Taking the tensor product with ξ_c and antisymmetrizing (or equivalently, taking the wedge product), the above equation is equivalent to

$$\xi_{[c} \nabla_a \xi_{b]} = \xi_{[c} \xi_a \tilde{\nu}_{b]} = 0, \quad (6)$$

This completes the proof. $\square$

From an operational point of view, staticity allows us to conveniently set up a coordinate system. Specifically, we may first choose an arbitrary coordinate system $\{x^\mu\}$ on Σ . Then, provided that ξ^a is nowhere vanishing on Σ , Thus, the coordinates $\{x^\mu\}$ can be naturally carried to a neighborhood of Σ via the isometry ϕ_t corresponding to ξ^a , and the parameter t of the map can then be used as a “time coordinate,” thereby establishing a complete four-dimensional spacetime coordinate system.

Define Σ_t as the set of points whose time coordinate has the value t . Let w^a be a tangent vector at a point $p \in \Sigma$. Then the corresponding vector to Σ_t at $\phi_t(p)$ is $\phi_{t*} w^a$. Let $C(s)$ be a curve in Σ with tangent w^a and $C(s_0) \equiv p$. Its image under ϕ_t lies in Σ_t , denoted by $\phi_t(C(s))$, with $\phi_t(C(s_0)) = \phi_t(p)$. For any scalar field $f \in \mathcal{F}_N$, we have

$$\begin{aligned} [(\phi_{t*} w^a)(f)]|_{\phi_t(C(s_0))} &= [(\phi_{t*} w^a)(f)]|_{\phi_t(p)} \\ &= [w(\phi_t^* f)]|_p = [w(\phi_t^* f)]|_{C(s_0)} = [w(\phi_t^* f)|_{C(s)}]|_{s=s_0} \end{aligned}$$

$$\begin{aligned}
&= [w(f|_{\phi_t[C(s)]})]|_{s=s_0} = w\{f[\phi(C(s))]\}|_{s=s_0} \\
&= \frac{d\{f \circ \phi[C(s)]\}}{ds} \Big|_{s=s_0}.
\end{aligned} \tag{7}$$

Therefore, $\phi_{t*}w^a$ is the tangent vector to $\phi_t(C(s))$, and hence is tangent to Σ_t . Moreover, since ϕ_t is an isometry, i.e., Eq. (1) holds, we have

$$0 = g_{ab}|_p w^a \xi^b = \phi_t^* g_{ab}|_p w^a \xi^b = g_{ab}|_{\phi_t(p)} (\phi_{t*}w)^a (\phi_{t*}\xi)^b. \tag{8}$$

This shows that ξ^a at $\phi_t(p)$ is orthogonal to the vector $\phi_{t*}w^a$ tangent to Σ_t . Since w^a was arbitrary, it follows that ξ^a is orthogonal to every Σ_t . This orthogonality guarantees the absence of spatiotemporal cross terms in the metric components. Since $g_{\mu\nu}$ is independent of t as well, the line element of the metric can ultimately be written as

$$\begin{aligned}
ds^2 &= g_{tt}(x^1, x^2, x^3) dt^2 + g_{\mu\nu}(x^1, x^2, x^3) dx^\mu dx^\nu \\
&= \xi^a \xi_a dt^2 + h_{\mu\nu}(x^1, x^2, x^3) dx^\mu dx^\nu \\
&= -V^2(x^1, x^2, x^3) dt^2 + h_{\mu\nu}(x^1, x^2, x^3) dx^\mu dx^\nu,
\end{aligned} \tag{9}$$

where the second equality uses $g_{\mu\nu} = (h_{\mu\nu} + \xi_\mu \xi_\nu)$ and the fact that ξ has no spatial components.

3 Spherically Symmetric Spacetimes

Definition 3.1. A spacetime is said to be spherically symmetric if its isometry group contains a subgroup isomorphic to the group $SO(3)$, and the orbits of this subgroup (i.e., the collection of points resulting from the action of the subgroup on a given point) are two-dimensional spheres.

Regarding the $SO(3)$ group and the construction of spherical coordinates, reference should be made to my notes of March [3]. In summary, in spherical coordinates $\{r, \theta, \phi\}$, the induced line element on the 2-sphere is

$$d\hat{s}^2 = r^2 (d\theta^2 + \sin^2 \theta d\phi^2). \tag{10}$$

Finally, choosing $\{t, r, \theta, \phi\}$ as spacetime coordinates, the line element of an arbitrary static, spherically symmetric metric can be written as

$$ds^2 = -f(r)dt^2 + h(r)dr^2 + r^2(d\theta^2 + \sin^2 \theta d\phi^2), \tag{11}$$

where the absence of cross terms between the radial and angular directions follows from the orthogonality of $(\partial/\partial r)^a$ with $(\partial/\partial \theta)^a$ and $(\partial/\partial \phi)^a$.

References

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- [2] Liang, C. B. and Zhou, B. *Introduction to Differential Geometry and General Relativity, Volume I*. 2nd ed. Beijing: Science Press, 2006.

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A Appendix

A.1 Report on Use of AI Tools

I primarily use AI for simple repetitive calculations, as well as for English translation and polishing. AI also assists me in explaining concepts, verifying my ideas, and providing valuable proof strategies.

A.2 What Is New in This Work

This text is my reading notes on Wald's book [1], intended to supplement the specific computational details, while following the basic content of the original work.

Specifically, in Sec. 1, I explain why Definition 1.1 reflects time translation symmetry, and provide a detailed analysis of why stationarity is equivalent to the existence of a Killing vector field.

In Sec. 2, I prove Theorem 2.2, restate the procedure for constructing the coordinate system in my own words, and prove that ξ^a is orthogonal to every Σ_t .

Since the content of Sec. 3 has already been discussed in detail in my previous note [3], I do not expand on it here.

Finally, it should be emphasized that this note is merely a set of reading notes on Wald's book; the overall logic and ideas are inherited from Wald's work. My contribution consists only of rephrasing, reorganizing, and supplementing the specific computational and proof details. If there is anything of value in this note, the credit should first go to Wald's book. Should the copyright holder believe that this note infringes upon the copyright of Wald's book, please contact me and I will remove it promptly.